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L-Lactide polymerization studied by ^1H NMR with diffusion-ordered spectroscopy (DOSY): a “One NMR Tube Experiment” providing data on monomer conversion, polymer structure, M_n and M_w †

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L-Lactide polymerization was studied by a “One NMR Tube Experiment”. Traditional ^1H NMR, which provides structural information, conversion values, and number average molar mass (M_n), was combined with diffusion-ordered spectroscopy (DOSY), yielding a mass averaged molar mass (M_w). Thus, a complete kinetic experiment could be performed without sampling, and hence in one NMR tube.

Diffusion-ordered spectroscopy (DOSY) was first described fifty years ago, and today every NMR spectrophotometer has the capability to conduct DOSY experiments.¹ DOSY gives the dependence of the intensity of the peak ascribed to given nuclei on their diffusion coefficient (D) and magnetic field gradient.¹ The physical principle of DOSY has been explained clearly in the literature and various aspects of the experimental set-up have been discussed.^{2–4} An exhaustive review of original papers devoted to the method and its applications in various areas of organic chemistry, including polymer chemistry, has also been published.⁵ In polymer chemistry, DOSY has mostly been used to determine M_w . We here describe our application of this method to the polymerization of L-lactide (LA), to complement our kinetic work.

The detailed description of the theoretical basis of the method, which is related to the Stokes–Einstein relationship (1) between the size of molecules and their diffusion coefficients, has been given in several papers.^{6–9}

$$D = \frac{Tk_B}{6\pi\eta R_H} \quad (1)$$

In this equation, T is the absolute temperature, k_B is the Boltzmann constant, η is the viscosity of the medium, and R_H is the hydrodynamic radius of the studied molecule.

On the basis of mean field theory, molar mass M is related to R_H and leads to the dependence of M on D according to eqn (2).

$$R_H = KM^\nu \quad (2)$$

In this simple equation, which resembles the Mark–Houwink–Sakurada equation,¹⁰ K is the proportionality constant and ν is the scaling exponent.

Finally, the relationship between M and the diffusion coefficient has a relatively simple form (3):

$$D = K'M^{-\nu} \quad (3)$$

The coefficients K' and ν have been determined for a few polymer-solvent systems.^{7,11–13}

Recently Grubbs *et al.* applied DOSY for the first time in studies of a controlled polymerization of styrene and of 1,5-cyclooctadiene (ROMP).¹³ The authors acquired the monomer conversion and mass average molar mass (M_w) from analyses of samples taken from the reaction mixture.

We decided to follow the approach used by Grubbs *et al.* in our study of the polymerization of L-lactide, although in place of sampling the reaction mixture we performed polymerization in a sealed NMR tube, acquiring all the information “on-line” without taking samples. This was possible because the rate of the studied process is sufficiently slow at the measurement temperature, assuring that neither the measurements nor the polymerization were disturbed.

This work was divided into three parts: first, the dependence of the diffusion coefficient on M_w for poly(L-lactide) (PLA) was established; then, this dependence was verified in a kinetics experiment involving sampling the polymerization mixture; and finally, the “One NMR Tube Kinetic Experiment” was carried out. All experimental details, analytical procedures, and results of the characterization of the synthesized samples are described in ESI.†

Five samples of PLA were prepared, by polymerization with an $\text{Sn}(\text{2-ethylhexanoate})_2$ ($\text{Sn}(\text{Oct})_2$)/benzyl alcohol (BzOH) catalytic system, and precipitated from the polymerization solution. M_w values were measured by size exclusion chromatography.

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Table 1 Characteristics of PLA samples

Id	$M_n(\text{th})^a$	M_n^b	M_n^c	M_w^c	$\bar{D}^c$	$10^{11} \times D^d (\text{m}^2 \text{s}^{-1})$	M_w^d	Error ^e (%)
A	3200	4000	4100	4200	1.03	20.301	4207	−0.6
B	4660	6450	6300	6700	1.05	14.977	6601	3.5
C	10 500	13 100	13 500	13 900	1.03	9.857	13 861	0.2
D	20 100	20 100	21 000	21 600	1.03	7.301	22 065	6.3
E	29 700	30 100	31 000	33 700	1.09	5.838	33 056	−1.1

^aBased on stoichiometry and conversion. ^bIntegration of ^1H NMR spectra. ^cSEC with MALLS detection. ^dDOSY NMR in C_6D_6 . ^e $(M_w(\text{SEC}) - M_w(\text{DOSY})) \times 100/M_w(\text{SEC})$.

graphy (SEC) and D using DOSY. In addition, number average molar masses (M_n) were determined using ^1H NMR and SEC, and compared with M_n calculated from the conditions used for polymerization. The tabulated (Table 1) data were used to determine the analytical dependence of D on M_w (Fig. 1). The relative difference (called “error”) between M_w determined by SEC and M_w calculated from D , using the established analytical dependence, is given in the last column of Table 1.

As seen from the data presented in Table 1, the M_n determined using SEC with a MALLS detector and from the integration of signals in the ^1H NMR spectra are consistent with the theoretical values, based on the ratio $144.13 \times [\text{LA}]_0/[\text{BzOH}]_0$. Differences noted for the calculated and measured smallest masses could be ascribed to the removal of the low-mass molecules during precipitation of the crude samples. Low dispersities ($\bar{D}$), ranging from 1.03 to 1.09, indicate controlled behaviour of the applied polymerization process.

The DOSY spectra for PLA solutions were acquired according to the protocol used by Grubbs *et al.* (details given in ESI†).¹³ A novel ITAMeD algorithm and the Matlab® script provided recently by Urbańczyk *et al.* have been used for extraction of D values from attenuation decay curves.⁶ In contrast to other methods, ITAMeD does not require any assumptions regarding the composition of the analysed sample. The NMR

signal decay curves and obtained diffusion coefficient distributions are presented in the ESI.† The numerical values of D and M_w are given in Table 1.

Fig. 1 shows the dependence of the diffusion coefficients on the corresponding mass average molar masses. A straight line fits the experimental data with a correlation coefficient $r^2 = 0.9997$ and passes close to points obtained for benzene and LA. The proportionality constant K' was determined to be equal to $10^{-7.53(\pm 0.03)}$ and the value of the scaling exponent $\nu = 0.598(\pm 0.006)$. This scaling exponent value indicates that benzene is a good solvent for PLA, as it was established earlier on the basis of viscosity measurements.¹⁴ The D values extracted from the DOSY spectra using the T1/T2 TopSpin module are nearly the same as that obtained by using the ITAMeD method: $K' = 10^{-7.53(\pm 0.03)}$ and $\nu = 0.597(\pm 0.006)$ (cf. Fig. S5 in ESI†).

The M_w values, calculated from the analytical relationship (cf. Fig. 1), were then compared with the M_w values determined by SEC. The greatest difference that was noticed is 6.3%.

Even though the established dependence is valid only for the polymer concentration of 0.5 g l^{-1} , this dependence was successfully applied in studies of polymerization kinetics with different concentrations of the formed polymer by determining correction factors. Changes in the polymer concentration may also influence the diffusion coefficient by changing aggregation phenomena or simply by changing the viscosity of the solution. In order to correctly analyse the experimental data, the dependence of D on polymer concentration (from 0.5 to 30 g l^{-1}) was determined and plotted, as in Fig. 2.

These relationships were found to be linear, with their slopes not dependent on molar masses. The mean value of the slope (S) was calculated to be $(1.3 \pm 0.2) \times 10^{-12} \text{ m}^2 \text{ l g}^{-1} \text{ s}^{-1}$ and is the same for all molar masses within the studied range. Thus, the corrected value of the diffusion coefficient can be calculated from eqn (4).

$$D_{\text{corr}} = D_{\text{exp}} - S(c_{\text{exp}} - c_{\text{corr}}) \quad (4)$$

In this equation, c is concentration, and the indices “exp” and “corr” denote experimental and corrected values, respectively. Finally, the M_w values were calculated from the D_{corr} using the equation given in Fig. 1.

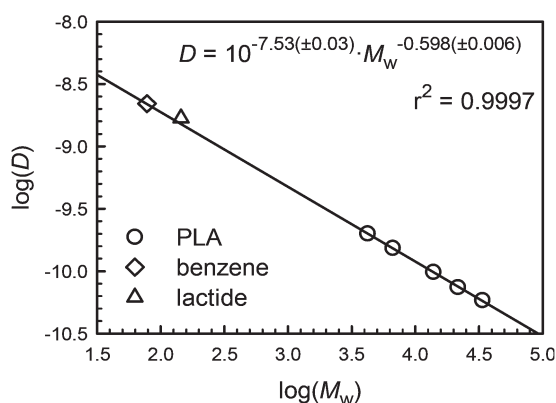


Fig. 1 Mass-averaged molar mass (M_w) vs. diffusion coefficient (D) log–log plot for PLA in benzene- d_6 .

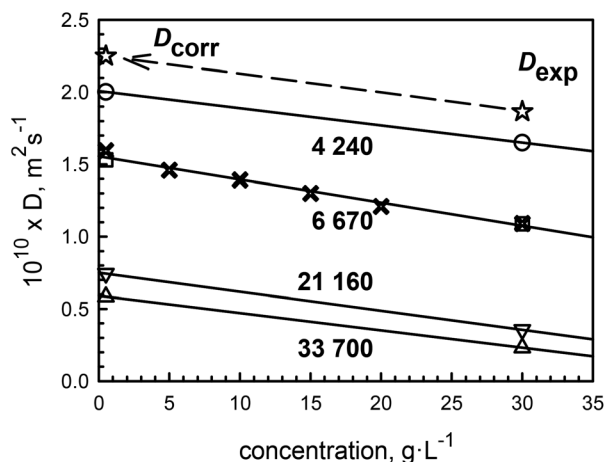


Fig. 2 Dependence of D on M_w and concentration of polymer. Open marks: data used for determining plot slopes. Crosses: data used for confirmation of linearity of the concentration dependence. Stars illustrate a correction introduced by eqn (4). Numbers indicate M_w values of the samples.

After having established the dependence of D on M_w and polymer concentration, it was necessary to verify the correctness of this relationship before performing the “One NMR Tube Experiment”. Therefore, a kinetic experiment on L-lactide polymerization was performed with sampling of the polymerization mixture (protocol and conditions of experiments are given in ESI†). The determined molar masses are given in Table 2.

The data in Table 2 (as well as Fig. S8 in ESI†) show, that for the dead macromolecules, both the DOSY and SEC methods give almost identical values of M_n and M_w . This observation additionally indicates that the relationship between M_w and D has been correctly determined. Moreover, D of polymer samples range from 1.00 to 1.07, as expected for chains formed in the controlled-living polymerization.

Application of DOSY for studies of the polymerization kinetics of L-lactide using the “One NMR Tube Experiment” approach is illustrated by polymerization conducted in sealed NMR tubes (Experimental details are given in ESI†). A polymerization mixture containing C_6D_6 , LA, and initiating/catalytic system $Sn(Oct)_2/BzOH$ was prepared and transferred into an NMR tube under vacuum. The tube was then sealed off and kept in a thermostat at 353 K. At predetermined time

Table 2 M_n , M_w , $\bar{D}$ and D as determined by SEC, 1H NMR and DOSY measurements in the kinetic experiment with sampling

Id	M_n^a	M_w^a	$\bar{D}^a$	M_n^b	M_w^c	$\bar{D}^d$	$10^{11} \times D^c$ ($m^2 s^{-1}$)
1	6500	6800	1.03	6800	6780	1.00	1.5042
2	8450	8600	1.02	8600	8620	1.01	1.3048
3	9400	9700	1.04	9800	10 100	1.03	1.1846
4	10 200	10 600	1.03	9400	10 100	1.07	1.1840

^a SEC with MALLS detection. ^b Integration of 1H NMR spectra. ^c From DOSY. ^d $M_n(^1H \text{ NMR})/M_w(\text{DOSY})$.

intervals, the tube was cooled and 1H NMR as well as DOSY spectra of the polymerizing mixture were acquired at 298 K. Polymerization practically does not take place at this temperature during the 20 minutes of the measurement.

The monomer conversion and degree of polymerization P_n were determined from integrations of the standard 1H NMR spectrum (Fig. 3). The spectrum shows the structure of active macromolecules being in equilibrium with the temporarily inactive chains with $-OH$ end groups, in agreement with ref. 15. The monomer conversion was calculated from the ratio of the integration of the signal from methine protons of the lactoyl units e (overlapped quartets at $\delta = 5.0\text{--}5.1$ ppm) to the integration of the monomer signal (quartet centred at $\delta = 3.75$ ppm). Signals e' and g broadened due to an equilibrium between active and inactive chains and therefore cannot be used for calculations of P_n of living macromolecules. P_n and M_n were calculated from the ratio of the integrations of signals e and d' (the latter at $\delta = 4.89$ ppm).

Fig. 4 shows the experimental plot of the left-hand side of eqn (5) as a function of time.

$$-\ln\left(1 - \frac{[I]_0 P_n}{[M]_0 - [M]_e}\right) = k_p [I]_0 t \quad (5)$$

The corresponding equation was derived for living irreversible polymerization in one of our previous works by combining $\ln([M]_0/[M]_e) = -k_p [I]_0 t$ and $P_n = ([M]_0 - [M]_e)/[I]_0$.¹⁶ eqn (5), after introduction of $[M]_e$, is valid for the reversible polymerization and can be used for determination of the propagation rate constant. The linearity of the plot (as in Fig. 4) means that termination and irreversible transfer are simultaneously

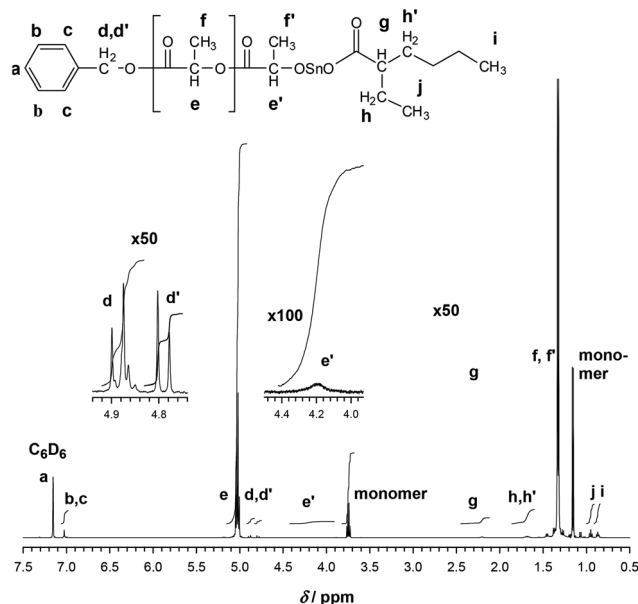


Fig. 3 1H NMR 500 MHz spectrum of the living PLA in C_6D_6 initiated by $Sn(Oct)_2/BzOH$ system. Temperature 353 K, reaction time 12 h, $[LA]_0 = 0.226 \text{ mol l}^{-1}$, $[BzOH]_0 = 2.99 \times 10^{-3} \text{ mol l}^{-1}$, $[Sn(Oct)_2]/[BzOH]_0 \approx 1.0$. Integrations (proton number): $d' = 1.0$, $e = 134.0$, monomer = 25.0.

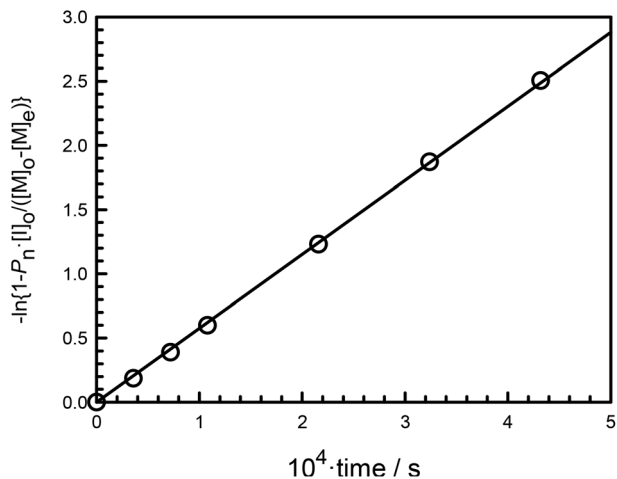


Fig. 4 Kinetics plot of the polymerization of LA in C_6D_6 initiated by $Sn(Oct)_2/BzOH$ system. Temperature 353 K, $[LA]_0 = 0.226 \text{ mol l}^{-1}$, $[BzOH]_0 = 2.99 \times 10^{-3} \text{ mol l}^{-1}$, $[Sn(Oct)_2]_0/[BzOH]_0 \approx 1.0$.

absent. $[I]_0$ in eqn (5) is equal to the invariant concentration of the growing chains. However, in the system studied in this work, the concentration of the growing chains, being invariant, is neither equal to $[Sn(Oct)_2]_0$ nor to $[BzOH]_0$. As the major aim of this work is to demonstrate the “One NMR Tube Experiment”, it suffices to estimate the $[I]_0$ value using an approach presented in the work from our laboratory as well as the kinetic studies by Morbidelli.^{17,18}

It was determined in our work that the instantaneous concentration of the active centres is a function of $[ROH]$ for an invariant $[Sn(Oct)_2]_0$.¹⁷ For the conditions used in the present work $[P^*] \approx [ROH]/3$ (where $[P^*]$ is the concentration of the momentarily active species). This approximate equality results from the formation of active species in the reversible reaction of $Sn(Oct)_2$ with $BzOH$, followed by the propagation step with a reversible interconversion of active chains into the temporarily inactive ones.

The slope in Fig. 4 ($k_p[I]_0$) is equal to $5.76 \times 10^{-5} \text{ s}^{-1}$ and it follows that $k_p \approx 0.058 \text{ l mol}^{-1} \text{ s}^{-1}$ ($[I]_0 = 1.134 \times 10^{-2} \text{ mol l}^{-1}$), and is close to $0.091 \text{ l mol}^{-1} \text{ s}^{-1}$, which was calculated by taking into account the temperature coefficient for k_p as determined by Morbidelli in bulk for the temperature range from 403 to 483 K.¹⁸

The diffusion coefficient values for the instantaneous polymer concentrations were extracted from the DOSY spectra acquired on-line and corrected to the proper concentration (eqn (4)). M_w values were obtained from the equation given in Fig. 1.

In Fig. 5, the dependence of M_n and that of M_w on conversion are shown. $\bar{D}$ values shown in the figure were calculated as equal to the ratio M_w/M_n .

The determined values of M_w (and therefore also $\bar{D}$) are higher than expected for controlled-living polymerization. There are several possible reasons: aggregation of the living chains, reversibility of propagation, particularly important at

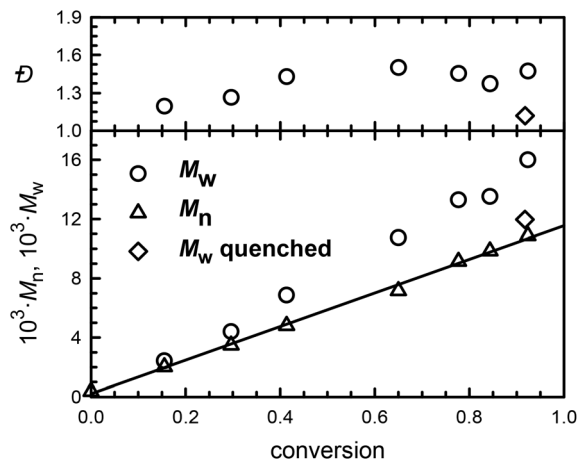


Fig. 5 Dependence of M_n , M_w and $\bar{D}$ on conversion in living polymerization of LA in C_6D_6 at 353 K, initiated by $Sn(Oct)_2/BzOH$. $[LA]_0 = 0.226 \text{ mol l}^{-1}$, $[BzOH]_0 = 2.99 \times 10^{-3} \text{ mol l}^{-1}$, $[Sn(Oct)_2]_0/[BzOH]_0 \approx 1.0$. Solid line is the least squares fit to the M_n data. Numerical data are shown in Table 3.

Table 3 Conversion, P_n , M_n , $\bar{D}$, M_w and $\bar{D}$ determined from 1H NMR and DOSY measurements for the “One NMR Tube Experiment”

Time (s)	Conv.	P_n^a	M_n^b	$10^{10} \times D \text{ m}^2 \text{ s}^{-1}$	M_w	$\bar{D}^c$
3600	0.115	11.5	2030	2.780	2430	1.20
7200	0.296	21.6	3490	1.947	4410	1.26
10 800	0.413	30.9	4820	1.494	6870	1.43
21 600	0.650	47.2	7200	1.144	10 700	1.49
32 400	0.777	61.0	9200	1.007	13 300	1.45
43 200	0.844	65.8	9900	0.9972	13 500	1.36
129 600	0.924	72.9	10 900	0.9017	16 000	1.47
129 600 ^d	0.918	71.5	10 700	1.072	12 000	1.12

^a Number of LA units. ^b $BzOH$ and $-SnOct$ end groups. ^c M_n/M_w . ^d Quenched.

higher conversion, and inter-conversion between the active and temporarily inactive chains. All these phenomena vanish for quenched samples. Indeed, M_w , and therefore also $\bar{D}$, drop to levels characteristic for macromolecules formed in the controlled-living conditions. Corresponding results obtained from 1H NMR/DOSY and SEC analyses of the “killed” polymers are shown in Table 2. A decrease of M_w and a decrease of $\bar{D}$ were noticed also after quenching the polymerization in the “One NMR Tube Experiment” (cf. Table 3 and Fig. 5). Hence, the described methodology also gives insight into the subtle phenomena in the polymerizing systems.

Conclusions

It has been shown, in the controlled polymerization of L-lactide, that using a combination of different NMR techniques (standard and DOSY) allows studying on-line the kinetics of polymerization together with the determination of M_n , M_w and $\bar{D}$.

Moreover, in the same experiment carried out in a single NMR tube, it was possible to analyze the structure of poly(L-lactide).

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